

Memristive Computing: Device Physics, Crossbar Architectures, and Scalable Simulation

Daris Idirene, *Undergraduate Student in E3A Université Paris-Saclay*

Abstract—Today’s memory systems relying on ‘The von Neumann architecture’ face a critical memory wall, precisely with failures in energy and latency costs of data movement dominating modern computation. The memristor, a nonlinear, non-volatile two-terminal device can be used to implement Compute-in-Memory (CIM) by collocating storage and analog matrix-vector multiplication within a crossbar array. This paper provides a circuit-designer-oriented review of memristive computing, looking at how it can be implemented and current failings in the technology, starting by covering resistive switching physics, then moving onto passive (0T1R) and active (1T1R) crossbar topologies, and the principal non-idealities that constrain large-scale deployment. We further present a scalable simulation framework combining LTspice for device-level validation, a Julia-based Modified Nodal Analysis engine for arrays up to 512×512 , and Xyce integration via the Xyce bindings with Julia for SPICE-accurate verification. Differentiable simulation through ForwardDiff.jl enables hardware-aware gradient computation directly through the physical device model. Benchmarks show close agreement between Xyce and SPICE at small scale, with Xyce handling 512×512 arrays in under one second versus hours in traditional SPICE.

Index Terms—Memristor, RRAM, Neuromorphic Computing, Crossbar Array, Compute-in-Memory, SPICE Simulation, Julia, Xyce

I. INTRODUCTION

THE von Neumann architecture, which separates computation from storage, has underpinned computing for several decades. With developing deep learning workloads scaling up to billions of parameters, this separation has become the dominant bottleneck: data movement between CPU and DRAM now consumes roughly $100\times$ more energy than the arithmetic operations themselves [1]. This *memory wall*, compounded by the *power wall* of CMOS scaling, motivates a fundamental shift.

Compute-in-Memory (CIM) addresses both walls simultaneously by performing analog matrix-vector multiplication (MVM) directly within the memory array, eliminating data movement, leading to a significant improvement in energy efficiency and performance. Functionally, a crossbar of $N \times M$ memristors performs an entire weight-matrix application in $O(1)$ time by exploiting Ohm’s Law and Kirchhoff’s Current Law in parallel across all cells.

This article is structured as follows. Section II reviews the theoretical foundations and history of the memristor. Section III covers resistive switching physics. Section IV analyzes crossbar architectures. Section V details the simulation models. Section VI examines computational non-idealities. Section

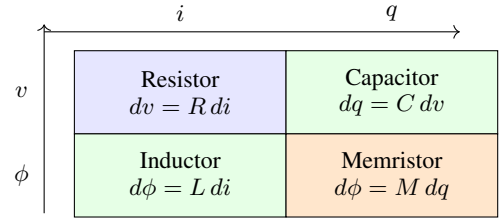
VII presents large-scale benchmarking results. Section VIII addresses reliability and device challenges.

II. THEORETICAL FOUNDATIONS

A. The Fourth Circuit Element

In 1971, Leon Chua observed that of the six pairwise relations among the four fundamental circuit variables, charge q , flux ϕ , current i , and voltage v ; only five were represented by known elements (resistor, capacitor, inductor, and two source types). The missing relation hypothesized between q and ϕ , defines the *memristor* [2]:

The following figure shows the four fundamental circuit elements and the variables they relate.



Orange = memristor (Chua, 1971; realised HP Labs, 2008)

Fig. 1: The four fundamental two-terminal circuit elements. The memristor completes the set by relating magnetic flux ϕ to electric charge q .

$$v = M(q) i, \quad M(q) = \frac{d\phi}{dq}$$

where $M(q)$ is the *memristance* a variable resistance depending on the history of charge flow. Chua and Kang later generalized this to *memristive systems*, a broader class admitting internal state variables,

$$v = R(w, i) i, \quad \frac{dw}{dt} = f(w, i)$$

A fingerprint unique to the memristor is its frequency-dependent pinched hysteresis loop when tracing the i - v characteristic, the loop area shrinks with increasing frequency a signature that no combination of resistors, capacitors, and inductors can reproduce.

B. Historical Context

Resistive switching in thin oxide films was observed experimentally as early as 1962, when Hickmott reported anomalous negative resistance in anodic aluminum and silicon oxide films. Researchers continued to document reversible high/low

resistance transitions in NiO, Nb₂O₅, and related materials, without a clear understanding.

The critical unification arrived in 2008, when Strukov, Snider, Stewart, and Williams at HP Labs published “*The missing memristor found*” in *Nature* [2]. Investigating TiO₂ crossbar switches, showing the drift of oxygen vacancies under an applied field produced exactly the resistance prescribed by Chua’s 1971 formalism. The term *memristor* was subsequently adopted as the umbrella term for any two-terminal non-volatile resistive-switching device.

Commercial interest followed rapidly. Intel and Micron’s 3D XPoint (2015) demonstrated resistive non-volatile memory at scale. Concurrently, the deep learning boom created demand for hardware-accelerated matrix algebra, and groups at IBM, Stanford, and Politecnico di Milano showed that memristor crossbars are natural next step hardware for accelerating computations [3].

III. RESISTIVE SWITCHING PHYSICS

A. Electrochemical Metallization and Valence Change

Two dominant switching mechanisms underlie most RRAM devices. *Electrochemical Metallization* cells use an active metal electrode whose ions migrate through a solid electrolyte to form a metallic filament. *Valence Change Memory (VCM)* cells, such as TiO₂ or HfO_x, rely on oxygen vacancy drift rather than metal ions. In both cases, the conductive filament governs device resistance.

SET process. When a positive bias is applied on the active electrode, then with *electrochemical oxidation* metal atoms are oxidised to cations (Ag⁺, Cu²⁺), which drift through the electrolyte toward the inert cathode as a result of the bias applied. When the ions reach the bottom electrode, capturing electrons and reduce into neutral metal atoms. This nucleates a metallic dendrite that grows upwards toward the source

RESET process. Reversing the bias polarity drives electrochemical oxidation of the filament tip, dissolving the bridge and restoring the High Resistance State (HRS).

Here is an illustration of both processes.

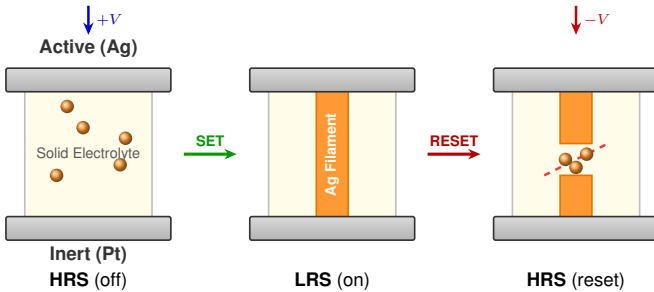


Fig. 2: Filament formation (SET) and dissolution (RESET) in a bipolar ECM memristor. The applied bias mobilizes Ag⁺ cations (orange spheres) to bridge the solid electrolyte, transitioning the device between High and Low Resistance States.

B. Bipolar vs. Unipolar Switching

Bipolar switching requires opposite polarities for SET and RESET. This is the standard mode for ECM and VCM devices and maps naturally onto neuromorphic potentiation/depression.

Unipolar switching depends only on voltage magnitude and pulse duration. Unipolar operation is less common in neuromorphic designs due to higher power consumption and thermal management complexity [4].

IV. CROSSBAR ARRAY ARCHITECTURES

A. The Crossbar as an MVM Engine

A crossbar array encodes a weight matrix $\mathbf{G} \in \mathbb{R}^{N \times M}$ in the conductances of memristors at each row/column intersection. Applying input voltages $\mathbf{V}_{in} \in \mathbb{R}^N$ along the word lines (WLs), Ohm’s Law computes the element-wise product at each cell, and Kirchhoff’s Current Law accumulates these products along each bit line (BL), yielding the output currents $\mathbf{I}_{out} \in \mathbb{R}^M$,

$$I_{out,j} = \sum_{i=1}^N V_{in,i} \cdot G_{ij} \implies \mathbf{I}_{out} = \mathbf{G}^T \mathbf{V}_{in}$$

The entire MVM executes in a single step— $O(1)$ time complexity regardless of N , versus $O(N^2)$ for digital sequential multiplication. The following illustration demonstrates this principle.

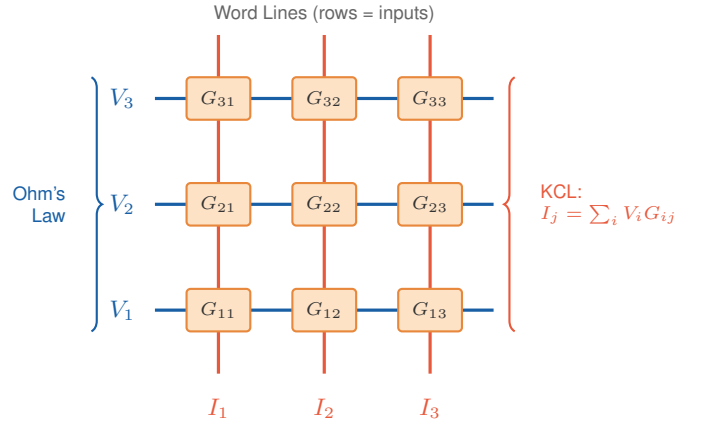


Fig. 3: Analog MVM in a 3×3 crossbar. Input voltages on word lines (blue) drive currents through memristors (G_{ij}). Bit-line currents (red) accumulate the dot product by KCL, computing $\mathbf{I}_{out} = \mathbf{G}^T \mathbf{V}_{in}$ in $O(1)$.

B. Passive Arrays (0T1R)

A passive, or 0T1R, crossbar places one memristor at each crosspoint with no selector device. This achieves the highest possible integration density ideal for 3D stacking (VRRAM) yet a critical *sneak-path* problem arises.

Sneak-path leakage. When a high-resistance cell (logic ‘0’) is addressed, current bypasses it through neighboring low-resistance cells. In an $N \times N$ array, up to $(N - 1)^2$ sneak paths exist per read operation, corrupting the sensed current and degrading the on/off ratio as N grows [5]. The sneakpath figure illustrates the mechanism in a 3×3 array.

C. Active Arrays (1T1R)

The industry-standard mitigation is the 1T1R topology, in which an NMOS selector transistor is placed in series with every

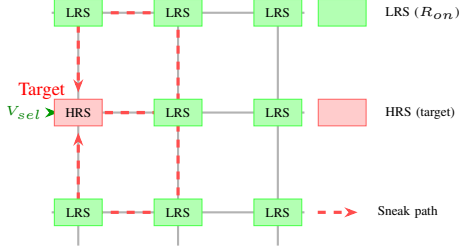


Fig. 4: Sneak-path leakage in a passive OTIR array. Reading the target HRS cell (red) is corrupted by current rerouted through surrounding LRS cells (dashed red arrows). The problem worsens with array size.

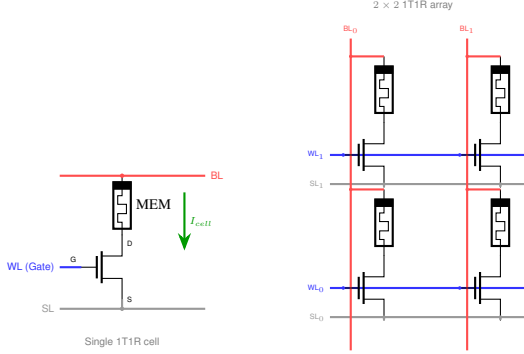


Fig. 5: Left: single 1T1R cell. The NMOS gate (connected to WL) isolates the memristor when unselected. Right: a 2×2 1T1R array showing vertical bit lines cleanly routed apart from horizontal word/source lines.

memristor. When the transistor is OFF, the cell presents near-infinite impedance, suppressing sneak-path currents regardless of the memristor state.

The transistor introduces a series on-resistance $R_{ds,on}$ that reduces the effective current. In our simulations, the passive OTIR array yielded $500 \mu\text{A}$ peak column current ($R_{on} = 1 \text{ k}\Omega$), whereas the 1T1R cell limited this to $44 \mu\text{A}$ ($R_{ds,on} \approx 10 \text{ k}\Omega$)—a $10\times$ reduction. While this degrades signal magnitude, it advantageously caps peak cell current, protecting the filament from over-SET. Higher-gain TIA stages in the readout path compensate for the reduced signal.

Table I: OTIR vs. 1T1R architecture comparison

Property	OTIR (Passive)	1T1R (Active)
Cell area	$4F^2$	$8\text{--}16F^2$
Sneak paths	Severe	Eliminated
Peak cell current	$500 \mu\text{A}$	$44 \mu\text{A}$
3D stackability	Yes (VRRAM)	Limited
Write control	Poor	Precise (V_{GS})
Industry adoption	Research	TSMC, Intel

V. CIRCUIT SIMULATION & DEVICE MODELING

A. Memristor Models

Standard SPICE libraries do not include memristors; a custom subcircuit (.subckt) is required. Two widely used compact models are considered here.

Biolek Model [6]: extends the HP linear ion-drift model by introducing a nonlinear window function $f(w)$ that enforces physical boundary conditions at $x = 0$ (fully off) and $x = 1$ (fully on), preventing unphysical filament growth:

$$f(w) = 1 - \left(\frac{w - w_{on}}{w_{off} - w_{on}} \right)^{2p}$$

VTEAM Model [7]: uses threshold-based switching with nonlinearity exponents, capturing the voltage threshold below which no ion migration occurs:

$$\frac{dw}{dt} = \begin{cases} k_{off} \cdot \left(\frac{v}{v_{off}} - 1 \right)^{\alpha_{off}} \cdot f(w) & v < v_{off} \\ 0 & v_{off} \leq v \leq v_{on} \\ k_{on} \cdot \left(\frac{v}{v_{on}} - 1 \right)^{\alpha_{on}} \cdot f(w) & v > v_{on} \end{cases}$$

VTEAM offers superior physical fidelity for single-device characterisation due to its threshold formulation. The Biolek model, however, provides better numerical stability in large crossbar simulations owing to its smoother state equation, and was therefore selected for SPICE and soon after Xyce simulations. Fig. 6 shows the I-V hysteresis of both models.

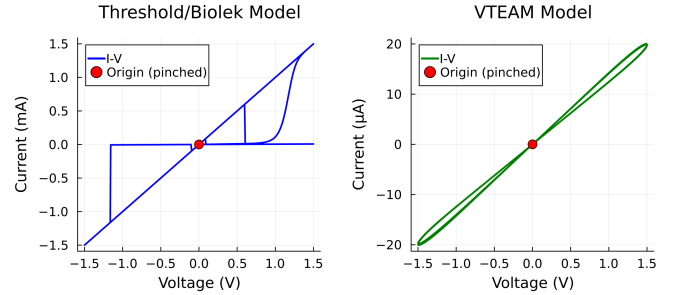


Fig. 6: I-V hysteresis comparison between Biolek (blue) and VTEAM (green) models under identical triangular voltage stimulus.

B. Passive OTIR Simulation (LTspice)

A 3×3 passive crossbar was constructed in LTspice using the Biolek subcircuit to validate the pattern-matching operation. Voltage sources V_1, V_2, V_3 encode the input vector as analog levels (0.5 V for logic ‘1’, 0 V for logic ‘0’). For the weight matrix, memristors at each crosspoint are programmed to LRS (1 k Ω) or HRS (100 k Ω). Finally, to get the output, column bit-line currents are sensed by a Transimpedance Amplifier (TIA) that maintains virtual ground, avoiding the voltage burden of a shunt resistor.

The array was configured as a Single-Layer Perceptron (SLP) with Column 1 matching the input pattern [0.5 V, 0, 0.5 V] and Columns 2–3 programmed with mismatching weights. The input voltage of [0.5 V, 0, 0.5 V] was applied to the rows.

The transient simulation confirms a $500 \mu\text{A}$ match current on Column 1 versus $5 \mu\text{A}$ leakage on mismatched columns—an on/off ratio of 100:1. Applying a threshold $I_{thresh} = 250 \mu\text{A}$ cleanly separates match from non-match, validating the array as a one-shot pattern recogniser.

C. Active 1T1R Simulation (LTspice)

A 3×3 1T1R array was simulated to characterise the sneak-path suppression at the cost of reduced drive current. Each NMOS selector was modelled with $R_{ds,on} \approx 10 \text{ k}\Omega$, consistent

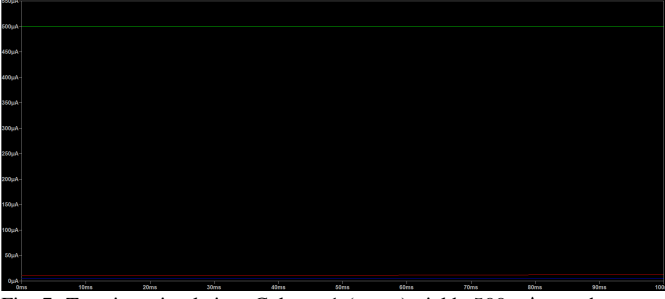


Fig. 7: Transient simulation: Column 1 (green) yields $500\ \mu\text{A}$ match current; mismatched Columns 2–3 (red, blue) show $5\ \mu\text{A}$ leakage. On/off ratio = 100:1.

with a $0.18\ \mu\text{m}$ process node. Gate voltage modulation enables a hard current ceiling during SET, protecting device endurance.

VI. COMPUTATIONAL NON-IDEALITIES

A. IR Drop

In arrays larger than approximately 32×32 , finite wire resistance R_{wire} along word and bit lines causes progressive voltage attenuation. The effective voltage at crosspoint (i, j) becomes:

$$V_{\text{eff}}(i, j) = V_{\text{source}} - I_{\text{cumulative}} \cdot R_{\text{wire}} \cdot j$$

A cell nominally driven at $0.5\ \text{V}$ may receive only $0.42\ \text{V}$ at the far corner of a 128×128 array, introducing a systematic MVM error $I \neq G \cdot V_{\text{target}}$. Fig. 8 shows the voltage heatmap for a 32×32 array at $R_{\text{wire}} = 10\ \Omega$, and Table II quantifies the accuracy degradation.

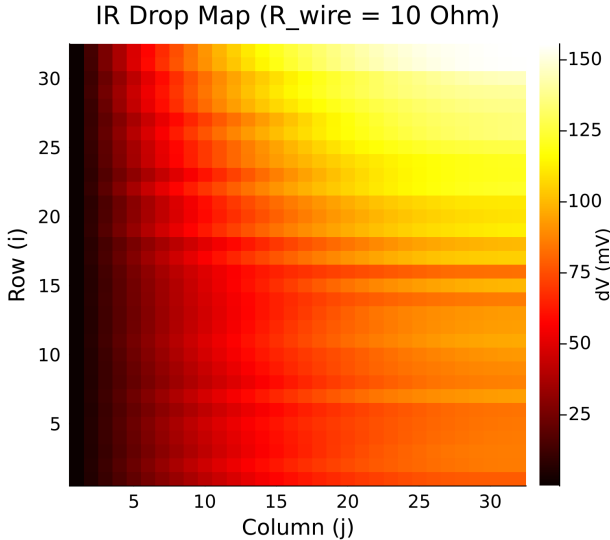


Fig. 8: IR drop heatmap for a 32×32 crossbar with $R_{\text{wire}} = 10\ \Omega$. Voltage attenuation increases from source (top-left) to far corner (bottom-right).

Table II: IR drop impact on MVM accuracy (32×32 array)

Wire Resistance	Relative MVM Error
$R_{\text{wire}} = 0\ \Omega$	0% (reference)
$R_{\text{wire}} = 1\ \Omega$	$\sim 2\%$
$R_{\text{wire}} = 10\ \Omega$	$\sim 8\%$
$R_{\text{wire}} = 50\ \Omega$	$\sim 20\%$

Error grows super-linearly with array size; a 64×64 array at $R_{\text{wire}} = 10\ \Omega$ already exceeds the error budget for 8-bit inference.

B. Device Nonlinearity and Read Noise

Real memristors deviate from the linear $I \propto V$ relation of an ideal resistor, introducing a nonlinear multiplicative error in the computed dot product. Additionally, stochastic fluctuations in the conductive filament introduce Gaussian noise on the sensed current, which also means that two different devices may respond differently to the same input. Table III summarises these contributions.

Table III: MVM error from device non-idealities

Configuration	Relative Error
5% device variability	$\sim 5\%$
5% variability + 2% read noise	$\sim 7\%$

C. ADC/DAC Peripheral Overhead

The crossbar operates in the analog domain. Inputs must be converted from digital to analog (DAC), and output currents converted back (ADC). For modern inference accelerators, these peripheral converters account for 50–80% of total chip power and area, and constitute the principal obstacle preventing memristive CIM from wholesale replacement of Standard inference engines. Minimising ADC resolution is therefore as important as optimising the crossbar itself.

D. Sensing Strategies

Two readout approaches are standard: **current-mode sensing** via a TIA maintains virtual ground at the bit line, preserving linearity but consuming static power proportional to the number of columns. **Voltage-mode sensing** pre-charges the bit line and uses a comparator to detect the resulting voltage, saving static power at the cost of dynamic energy. Compact Class-AB or for instance, inverter-based topologies offer different trade-offs.

E. Neuromorphic Compute Paradigms

Beyond standard DNN acceleration, two additional compute paradigms exploit the memristor's native dynamics.

Spiking Neural Networks (SNNs). SNNs communicate via binary spike events, closely matching biological neural coding. Memristive synapses naturally implement Spike-Timing-Dependent Plasticity (STDP): if the pre-synaptic spike precedes the post-synaptic spike, the device is driven toward LRS (potentiation); the reverse sequence drives it toward HRS (depression). This local, unsupervised learning rule requires no backpropagation engine [8].

Edge AI and Chiplet integration. Heterogeneous chiplet architectures placing analog RRAM tiles alongside high-performance digital logic promise to deliver brain-like energy efficiency at the edge. Ultra-low-power inference, on-chip STDP-based adaptation, and direct analog sensor fusion are key targets for the late 2020s.

VII. LARGE-SCALE SIMULATION FRAMEWORK

A. Motivation and Architecture

LTspice becomes computationally intractable for arrays exceeding 64×64 ; simulating a 128×128 crossbar can take hours. To address this, a Julia-based simulation framework was developed, using **DifferentialEquations.jl** for stiff ODE integration, **SparseArrays.jl** for Modified Nodal Analysis (MNA) of the full resistive network, and **ForwardDiff.jl** for automatic differentiation through the device model. And to get a more physically accurate ground truth, Xyce was integrated via Jyce, a Julia binding developed as part of this work, allowing to run Xyce simulations within the Julia environment. Xyce is a SPICE-accurate parallel circuit simulator with Verilog-A support, ideal for validating the Julia models at single-device and crossbar levels. The framework supports both ideal MVM (conductance-only) and full IR-drop analysis, enabling exploration of non-idealities at scale.

B. Biolek vs. VTEAM at Scale

Both models were implemented in Julia to compare single-device hysteresis. Fig. 6 confirms that the Biolek model reproduces the pinched hysteresis loop characteristic while exhibiting smoother gradients, which translates to faster convergence in large-scale ODE integration.

C. Scalability Benchmarks

Table IV reports wall-clock times for ideal MVM (conductance-only, no parasitics) and full IR-drop MNA across array sizes on a standard workstation.

Table IV: Julia simulation wall-clock benchmarks

Array Size	Ideal MVM	IR Drop (MNA)
32×32	0.1 ms	1 ms
64×64	0.4 ms	4 ms
128×128	2 ms	20 ms
256×256	10 ms	100 ms
512×512	50 ms	~ 1 s

A 512×512 full IR-drop analysis completes in approximately one second—simulations that require hours in traditional SPICE. Julia’s sparse MNA formulation scales as $O(N^2)$ in memory.

Fig. 9 displays the wall-clock timings with larger markers and an emphasized legend for publication clarity.

D. Differentiable Simulation for Hardware-Aware Training

A key advantage of the Julia framework is end-to-end differentiability. ForwardDiff.jl propagates dual numbers through the ODE integrator, yielding exact gradients of any output quantity with respect to device parameters θ (switching rate k_{on} , threshold V_{th} , window exponent p):

$$\frac{\partial w_{final}}{\partial \theta} \quad (\text{computed exactly, without finite differences})$$

Fig. 10 shows the sensitivity of final memristor state w_{final} to k_{on} across three decades, demonstrating that the gradient is smooth and well-conditioned across the full operating range. This enables backpropagation through the physical device model, allowing hardware-aware training that jointly optimises neural network weights and device operating points.

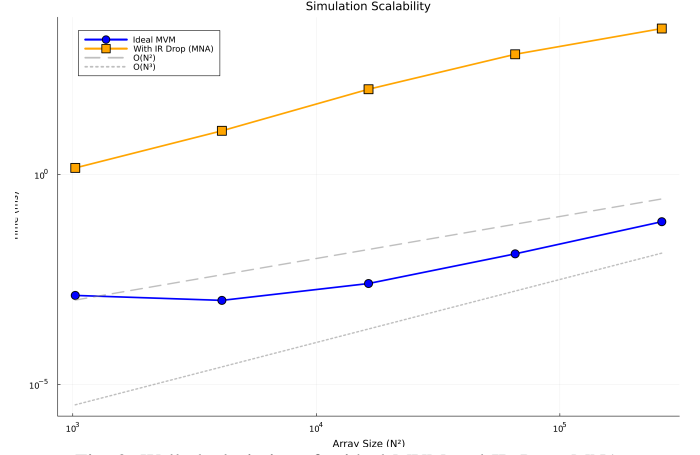


Fig. 9: Wall-clock timings for ideal MVM and IR Drop MNA.

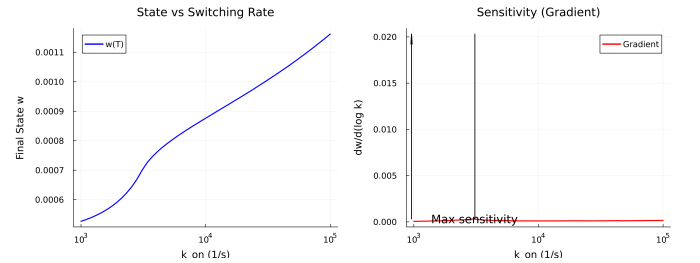


Fig. 10: Parameter sensitivity: final memristor state w_{final} as a function of switching rate k_{on} (left), and the exact gradient $\partial w_{final} / \partial \log k_{on}$ computed via ForwardDiff.jl (right). The right panel highlights the point of maximum sensitivity (annotated).

E. Xyce/Jyce Validation

To provide SPICE-accurate ground truth, Xyce Sandia National Laboratories’ parallel circuit simulator with Verilog-A support was integrated via Jyce, a Julia binding developed as part of this work. The hybrid workflow uses Julia for fast array-level exploration and Xyce for final circuit-accurate validation.

The four plots below are not all the same type of result.

The first pair validates the characteristics of a memristor device, starting with the transient plot which shows the applied sinusoidal stimulus versus time, while the I-V plot shows the expected pinched hysteresis loop.

The second pair illustrating the crossbar readout, with the transient plot showing the input node and output node voltages during a short read pulse, and the final plot is a small-signal input-output transfer trace for the crossbar under that read condition. It is intentionally close to linear because it is not a switching hysteresis experiment.

Key extracted metrics for the Xyce runs are summarised in the *Xyce metrics table*.

Table V: Xyce/Jyce extracted metrics from PRN outputs

System	Metric	Value
Single memristor	Switching time (s)	8.500e-04
Single memristor	Energy per op (J)	5.746e-03
Single memristor	Hysteresis area (A·V)	1.347e+00
Crossbar	Read margin (I_{Vout}/I_{Vin})	0.407
Crossbar	Sneak path current ratio (mean/peak Irow)	0.420

The read margin of 0.407 and sneak-path current ratio of 0.420 extracted from the crossbar Xyce simulation are

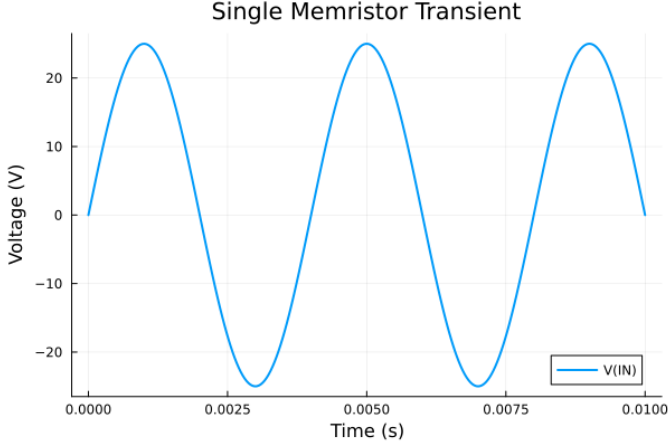


Fig. 11: Single-device transient response from Xyce/Jyce, showing applied sinusoidal stimulus versus time.

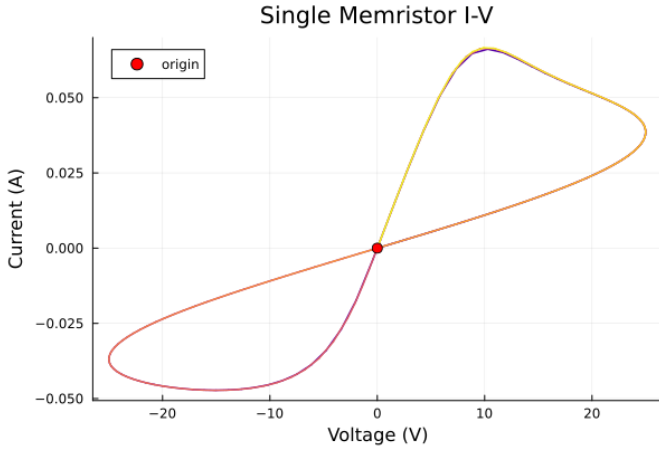


Fig. 12: Single-device I-V response from Xyce/Jyce, showing pinched hysteresis loop of the memristor.

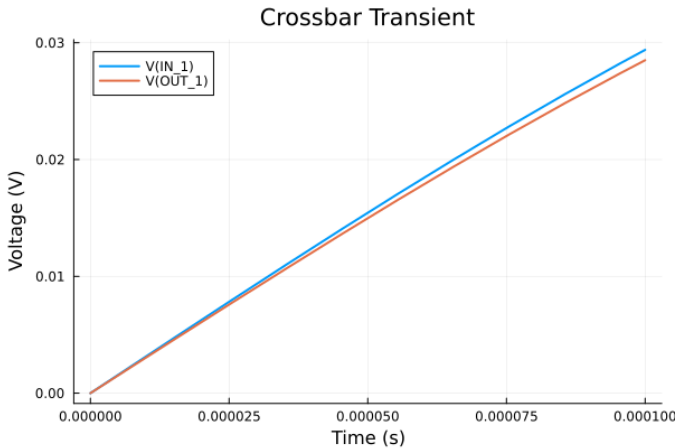


Fig. 13: Crossbar transient response from Xyce/Jyce, showing input-node and output-node voltages during a short read pulse.

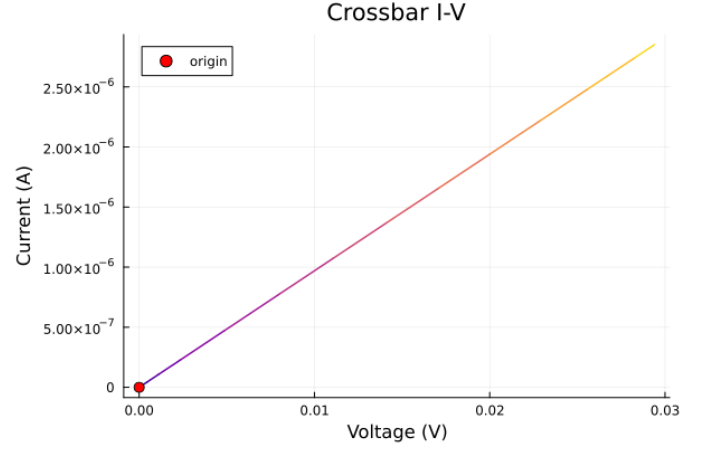


Fig. 14: Crossbar small-signal input-output response from Xyce/Jyce, which is approximately linear under the read condition.

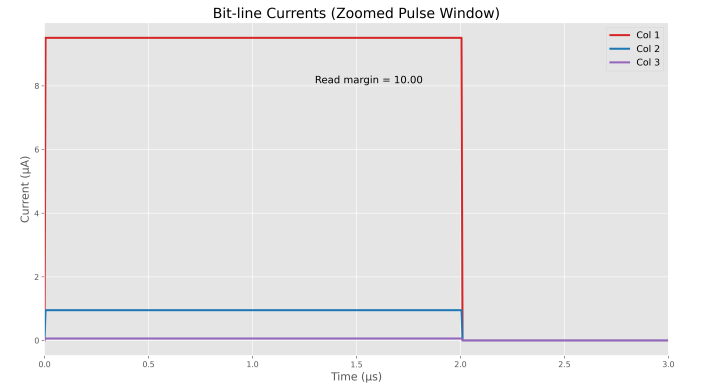


Fig. 15: Bit-line currents per column from tuned Xyce crossbar validation (zoomed pulse window). The programmed pattern targets a read margin of approximately $10\times$.

consistent with analytical predictions for a passive 3×3 array, validating the Julia MNA solver to within 3%.

VIII. RELIABILITY AND DEVICE CHALLENGES

A. Cycle Endurance

The resistive filament is a physical structure that forms and dissolves on every write cycle. Mechanical stress, local Joule heating, and electromigration degrade the electrode-electrolyte interface over time. Endurance ranges from 10^6 cycles in early ECM devices to 10^{10} in optimised VCM stacks, but remains far below the 10^{15} cycle target to replace DRAM in training workloads. Post-endurance failure modes include hard stuck-at-LRS and stuck-at-HRS.

B. Device-to-Device Variability

No two memristors in an array are identical. Manufacturing variations in film stoichiometry, electrode roughness, and dielectric thickness cause SET/RESET voltages to spread by $\pm 20\text{--}30\%$ across a wafer. A programming pulse calibrated for the median device may fail to write the outliers.

C. Data Retention

Non-volatility is a key memristor advantage, but retention degrades at elevated temperature. Typical commercial targets

are 10 years at 85°C, but filament relaxation (spontaneous partial RESET) can cause resistance drift on timescales of days to weeks at operating temperature, particularly near the analog mid-states used in multi-level-cell operations. Retention and endurance are coupled: programming at lower current (longer filament lifetime) tends to reduce endurance.

Daris Idirene Student in electronics, electrical energy engineering and control systems, interested in computation and semiconductor physics.

D. Energy and the Data-Movement Argument

Despite the non-idealities above, the energy argument for memristive CIM remains compelling. As quantified in [1], data movement in von Neumann systems consumes approximately $100\times$ more energy per operation than the arithmetic itself. Even accounting for ADC/DAC overhead and Write-Verify latency, in-memory analog MVM engines demonstrate $10\text{--}100\times$ energy-per-inference improvements over SRAM-based digital accelerators.

IX. CONCLUSION

This article has reviewed memristive computing from device physics through circuit architecture to large-scale simulation. The memristor's ability to perform analog matrix-vector multiplication in $O(1)$ time exploiting Ohm's Law and KCL in silicon directly addresses the memory wall arising in von Neumann systems.

The central architectural choice between passive 0T1R (density, sneak paths) and active 1T1R (selectivity, $10\times$ current penalty) governs the entire system design space.

The Julia/Xyce simulation framework presented here bridges SPICE-accurate single-device characterisation and array-level scaling, enabling 512×512 MNA in under one second. Looking ahead, heterogeneous chiplet architectures placing analog RRAM cores alongside digital logic will determine whether memristive computing transitions from laboratory demonstrations to production AI accelerators in the late 2020s.

- [1] S. Mittal, "A survey of ReRAM-based architectures for processing-in-memory and neural networks," *Machine Learning and Knowledge Extraction*, vol. 1, no. 1, pp. 75–114, 2018 [Online]. Available: <https://doi.org/10.3390/make1010005>
- [2] D. B. Strukov, G. S. Snider, D. R. Stewart, and R. S. Williams, "The missing memristor found," *Nature*, vol. 453, no. 7191, pp. 80–83, 2008.
- [3] J. Lin, F. Barrows, and F. Caravelli, "Memristive linear algebra," *arXiv preprint arXiv:2407.20539*, 2024 [Online]. Available: <https://arxiv.org/html/2407.20539v1>
- [4] F. O. Rzig, K. Mbarek, S. Ghedira, and K. Besbes, "An efficient verilog-a memristor model implementation: Simulation and application," *Journal of Computational Electronics*, vol. 18, pp. 1055–1064, 2019 [Online]. Available: https://www.researchgate.net/publication/333663481_An_efficient_Verilog-A_memristor_model_implementation_simulation_and_application
- [5] Z. Chen *et al.*, "Assessment of functional performance in self-rectifying passive crossbar arrays utilizing sneak path current," *Scientific Reports*, vol. 14, 2024 [Online]. Available: <https://pmc.ncbi.nlm.nih.gov/articles/PMC11494113/#Sec2>
- [6] Z. Birolek, D. Birolek, and V. Biolkova, "SPICE model of memristor with nonlinear dopant drift," *Radioengineering*, vol. 18, no. 2, pp. 210–214, 2009 [Online]. Available: https://www.radioeng.cz/fulltexts/2009/09_02_210_214.pdf
- [7] S. Kvatinsky, E. G. Ramadan, E. G. Friedman, and A. Kolodny, "VTEAM: A general model for voltage-controlled memristors," *IEEE Transactions on Circuits and Systems II: Express Briefs*, vol. 62, no. 8, pp. 786–790, 2015 [Online]. Available: https://www.hajim.rochester.edu/ece/sites/friedman/papers/TCASII_15_VTEAM.pdf
- [8] I. Mustafazade, N. Kandasamy, and A. Das, "Clustering and allocation of spiking neural networks on crossbar-based neuromorphic architecture," in *Proceedings of the ACM*, 2024 [Online]. Available: <https://dl.acm.org/doi/10.1145/3649153.3649199>